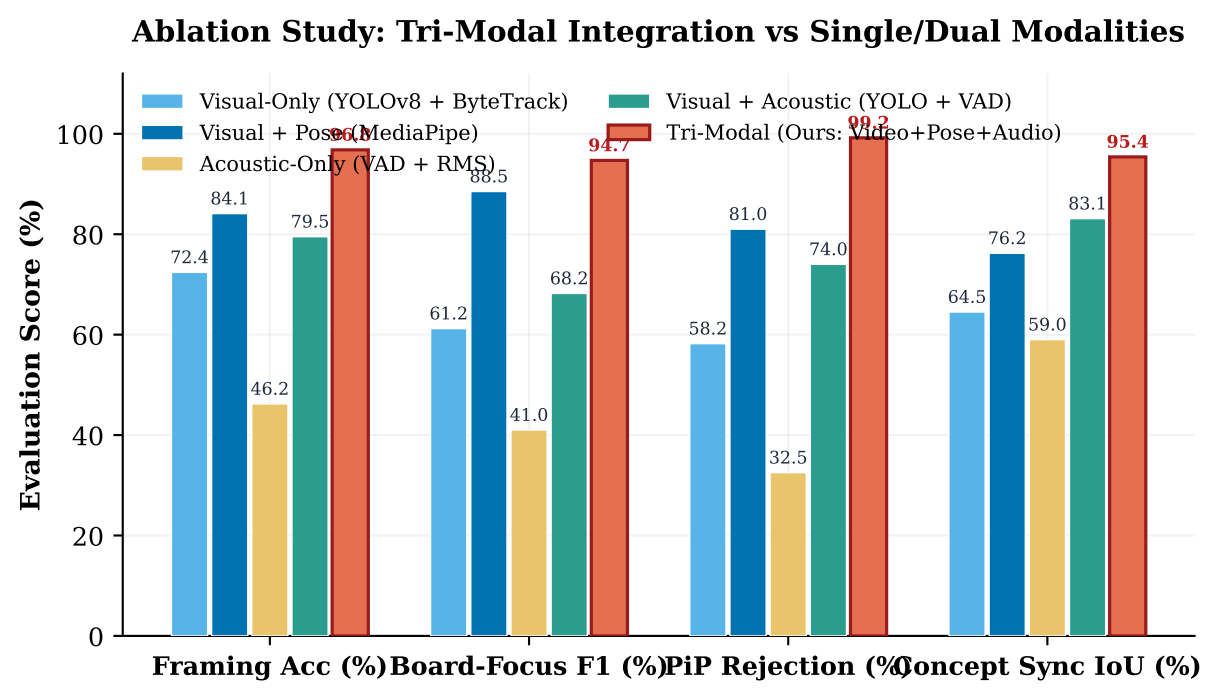
Ablation Study: Tri-Modal Integration vs Single/Dual Modalities
Visual-Only (YOLOv8 + ByteTrack)
Visual + Pose (MediaPipe)
Acoustic-Only (VAD + RMS)
Visual + Acoustic (YOLO + VAD)
Tri-Modal (Ours: Video+Pose+Audio)
Evaluation Score (%)
0
20
40
60
80
100
72.4
84.1
46.2
79.5
96.8
61.2
88.5
41.0
68.2
94.7
58.2
81.0
32.5
74.0
99.2
64.5
76.2
59.0
83.1
95.4
Framing Acc (%)
Board-Focus F1 (%)
PiP Rejection (%)
Concept Sync IoU (%)